Dietary BMAA Exposure in an Amyotrophic Lateral Sclerosis Cluster from Southern France

Background Dietary exposure to the cyanotoxin BMAA is suspected to be the cause of amyotrophic lateral sclerosis in the Western Pacific Islands. In Europe and North America, this toxin has been identified in the marine environment of amyotrophic lateral sclerosis clusters but, to date, only few dietary exposures have been described. Objectives We aimed at identifying cluster(s) of amyotrophic lateral sclerosis in the Hérault district, a coastal district from Southern France, and to search, in the identified area(s), for the existence of a potential dietary source of BMAA. Methods A spatio-temporal cluster analysis was performed in the district, considering all incident amyotrophic lateral sclerosis cases identified from 1994 to 2009 by our expert center. We investigated the cluster area with serial collections of oysters and mussels that were subsequently analyzed blind for BMAA concentrations. Results We found one significant amyotrophic lateral sclerosis cluster ($p = 0.0024$), surrounding the Thau lagoon, the most important area of shellfish production and consumption along the French Mediterranean coast. BMAA was identified in mussels (1.8 µg/g to 6.0 µg/g) and oysters (0.6 µg/g to 1.6 µg/g). The highest concentrations of BMAA were measured during summer when the highest picocyanobacteria abundances were recorded. Conclusions While it is not possible to ascertain a direct link between shellfish consumption and the existence of this ALS cluster, these results add new data to the potential association of BMAA with sporadic amyotrophic lateral sclerosis, one of the most severe neurodegenerative disorder.